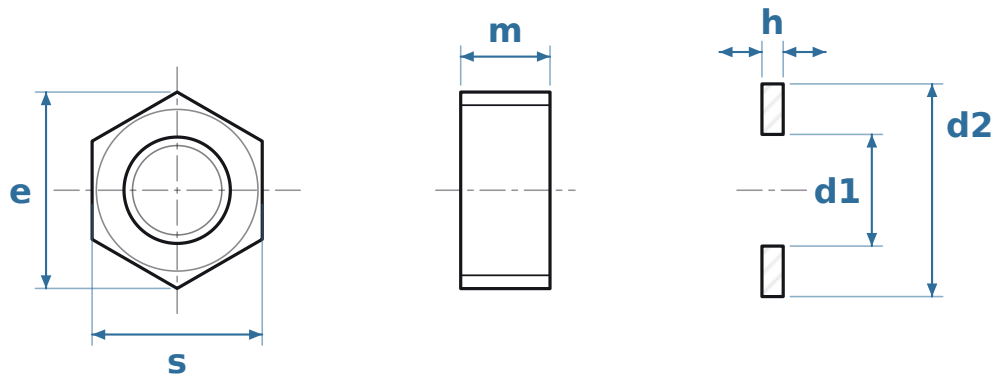
PRINTABLE REFERENCE CHART

Hex nut ISO 4032 and flat washer ISO 7089 dimensions

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1. ISO 4032 hex nut dimensions



nut, front view | nut, side view | washer, section

Regular hex nuts (style 1), coarse metric thread. Dimensions in mm: s is the spanner size (width across flats), m the height, e the minimum width across corners. The most common sizes are in bold.

Thread	Pitch P	Spanner s	Height m max	m min	e min
M1.6	0.35	3.2	1.3	1.05	3.41
M2	0.4	4	1.6	1.35	4.32
M2.5	0.45	5	2	1.75	5.45
M3	0.5	**5.5**	**2.4**	2.15	6.01
M4	0.7	**7**	**3.2**	2.9	7.66
M5	0.8	**8**	**4.7**	4.5	8.79
M6	1	**10**	**5.2**	4.9	11.05
M8	1.25	**13**	**6.8**	6.44	14.38
M10	1.5	**16**	**8.4**	8.04	17.77
M12	1.75	**18**	**10.8**	10.37	20.03
M14	2	21	12.8	12.1	23.36
M16	2	**24**	**14.8**	14.1	26.75
M18	2.5	27	15.8	15.1	29.56
M20	2.5	**30**	**18**	16.9	32.95
M22	2.5	34	19.4	18.1	37.29
M24	3	**36**	**21.5**	20.2	39.55
M27	3	41	23.8	22.5	45.20
M30	3.5	46	25.6	24.3	50.85
M33	3.5	50	28.7	27.4	55.37
M36	4	55	31	29.4	60.79

For the bolt clearance hole and the tap drill, see the metric thread chart; for tightening, the bolt torque chart.

2. ISO 7089 flat washer dimensions

Plain washers, normal series, product grade A, for bolts and nuts. Dimensions in mm: d1 is the hole diameter (minimum), d2 the outside diameter (maximum), h the nominal thickness and its range.

For bolt	Hole d1	Outside d2	Thickness h	h min … max
M1.6	1.7	4	0.3	0.25 … 0.35
M2	2.2	5	0.3	0.25 … 0.35
M2.5	2.7	6	0.5	0.45 … 0.55
M3	**3.2**	**7**	**0.5**	0.45 … 0.55
M4	**4.3**	**9**	**0.8**	0.7 … 0.9
M5	**5.3**	**10**	**1**	0.9 … 1.1
M6	**6.4**	**12**	**1.6**	1.4 … 1.8
M8	**8.4**	**16**	**1.6**	1.4 … 1.8
M10	**10.5**	**20**	**2**	1.8 … 2.2
M12	**13**	**24**	**2.5**	2.3 … 2.7
M14	15	28	2.5	2.3 … 2.7
M16	**17**	**30**	**3**	2.7 … 3.3
M18	19	34	3	2.7 … 3.3
M20	**21**	**37**	**3**	2.7 … 3.3
M22	23	39	3	2.7 … 3.3
M24	**25**	**44**	**4**	3.7 … 4.3
M27	28	50	4	3.7 … 4.3
M30	31	56	4	3.7 … 4.3
M33	34	60	5	4.4 … 5.6
M36	37	66	5	4.4 … 5.6

3. ISO 4032 vs DIN 934: the spanner sizes that change

DIN 934 was replaced by ISO 4032, but both are still found in stock. Only four sizes change spanner size (mm):

Thread	DIN 934 spanner	ISO 4032 spanner
M10	17	**16**
M12	19	**18**
M14	22	**21**
M22	32	**34**

Reference values taken from the article https://www.baouch.fr/en/blog/ecrou-rondelle-iso-4032-7089-dimensions/, which explains the assumptions, formulas and edge cases. For critical use, check the current standard or the manufacturer's documentation. Free to print and share with credit to the source.